

SEUPD@CLEF: Team retrix on Source Retrieval for Scientific Web Claims

Fabio Baldan¹, Davide Donati¹, Alvisè Garberino¹, Giancarlo Padoan¹ and Marco Tessari¹

¹University of Padua, Italy

Abstract

We present the results of the retrix group for the CheckThat! 2026 - Task 1, in which the goal is to identify the scientific paper implicitly referenced in a social media post. The input consists of short, noisy and usually informal posts that mention scientific findings without providing explicit links or explicit citations.

Our approach is designed to operate in a multilingual setting, as the dataset includes posts in English, German, and French. To handle this, we used multi-lingual language models.

DA FINIRE CON PIPELINE FINALEEEEEEE DA FINIRE CON PIPELINE FINALEEEEEEE DA FINIRE CON PIPELINE FINALEEEEEEE DA FINIRE CON PIPELINE FINALEEEEEEE

Keywords

Fact Checking, Information Retrieval, Scientific Document Retrieval, Social media fact verification, CLEF, Neural Re-ranking, Cross Encoder

1. Introduction

The rapid spread of scientific claims through social media creates a difficult fact-checking scenario: posts often report, simplify, or sensationalize research findings while omitting the original source. In many cases, the cited study is not linked explicitly with a URL or a bibliographic reference, but only mentioned indirectly through fragments such as the reported finding, the venue, the institution, or the authors. As a consequence, verifying such claims requires first identifying the scientific publication that is being implicitly referenced.

This problem is the focus of Task 1 of the CLEF 2026 CheckThat! Lab, *Source Retrieval for Scientific Web Claims*. Given a social media post containing a scientific claim and an implicit reference to a paper, the goal is to retrieve the correct publication from a fixed collection of 10.000 candidate papers written in English.

The task is challenging for several reasons. First, the linguistic gap between social media and scientific writing is substantial: posts are short, informal and often contain paraphrases, slogans, hashtags or incomplete statements, whereas scientific papers use more precise and technical language. Second, the reference to the source paper is implicit rather than explicit so the retrieval system must infer the connection from partial semantic evidence rather than from direct citations. Third, relevant clues may appear in different forms, such as mentions of a scientific result, a medical outcome, a research institution, or a publication venue, requiring the system to combine lexical and semantic evidence effectively.

An additional challenge is multilinguality: the official query sets include English, German and French posts while the candidate pool remains shared across languages.

Team Retrix approaches this problem with a three-stage sequential pipeline.

1. Raw tweet-style queries are enriched via pseudo-relevance feedback: BAAI/bge-large-en-v1.5 encodes both queries and corpus documents into a shared embedding space, retrieves the top-5

“Search Engines”, course at the master degree in “Computer Engineering”, Department of Information Engineering, and at the master degree in “Data Science”, Department of Mathematics “Tullio Levi-Civita”, University of Padua, Italy. Academic Year 2025/2026

✉ fabio.baldan.1@studenti.unipd.it (F. Baldan); davide.donati@studenti.unipd.it (D. Donati);
alvisè.garberino@studenti.unipd.it (A. Garberino); giancarlo.padoan@studenti.unipd.it (G. Padoan);
marco.tessari.8@studenti.unipd.it (M. Tessari)



© 2026 Copyright for this paper by its authors.
Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0).

semantically similar papers per query and extracts the most frequent terms from those papers as expansion tokens. Not-english queries are processed with BAAI/bge-m3 with the same high-level principles. Not-english queries are translated in English and then processed like the english ones.

2. The expanded queries are then fed into BAAI/bge-m3 operating in hybrid mode, which combines dense inner-product search via FAISS [1] with sparse lexical matching and ColBERT multi-vector scoring to produce an initial ranked list of candidates. The goal here is to have the highest recall possible.
3. Finally, the top candidates are reranked by a neural cross-encoder, in our case nvidia/llama-nemotron-rerank-1b-v2. in our case nvidia/llama-nemotron-rerank-1b-v2 fine-tuned on XXXXXXXX in our case answerdotai/ModernBERT-large fine-tuned on XXXXXXXXXX

Throughout the project we also investigated the effect of fine-tuning the reranker on domain-specific hard negatives extracted from the training set.

Evaluated on the CheckThat! 2026 Task 1 dataset our system ranked XXXXXXXXXXXX on the official leaderboard attaining an average Mean Reciprocal Rank at 5 (MRR@5) of XXXXXXXXXXXX % on the development set (XXXXXX place).

The paper is organized as follows: Section 2 discusses the most relevant research directions related to this task; Section 3 describes our approach; Section 4 explains our experimental setup; Section 5 discusses our main findings; finally, Section 6 draws some conclusions and outlooks for future work.

2. Related Work

The task addressed in this paper lies at the intersection of fact-checking and information retrieval. A natural starting point is lexical retrieval, where probabilistic ranking models such as BM25 remain strong baselines thanks to their efficiency, interpretability and robustness in candidate generation [2]. In the setting of scientific claim source retrieval, lexical matching is especially useful when a post mentions recognizable entities or terminology that also appears in the title, abstract, venue or author list of the referenced paper.

Neural retrieval methods have been proposed to mitigate the lexical mismatch between queries and documents by learning dense semantic representations. More broadly, neural information retrieval has shifted attention from exact token overlap to representation learning and learned matching functions [3]. In particular, dense dual-encoder architectures such as Dense Passage Retrieval (DPR) demonstrated that learned query and document embeddings can be highly effective for first-stage retrieval [4]. This line of work is especially relevant to our task because social media posts often paraphrase scientific findings instead of reusing the exact wording found in the original publication.

Between initial retrieval and final ranking, many modern systems adopt multi-stage pipelines that trade off efficiency and effectiveness. Late-interaction models such as ColBERT preserve fine-grained token-level matching while remaining substantially more scalable than fully pairwise neural scoring over the whole collection [5]. Likewise, transformer-based rerankers have become a common second-stage component for refining an initial candidate set with stronger query-document interactions [6]. These approaches motivate retrieval architectures that first generate a manageable set of candidates and then apply more expensive relevance estimation only where it is most useful.

Se fine tuning va bene è da aggiungere parte su hard negatives e ricerca embedding incrociati usate:
<https://arxiv.org/html/2505.18366v1> <https://huggingface.co/blog/train-reranker>

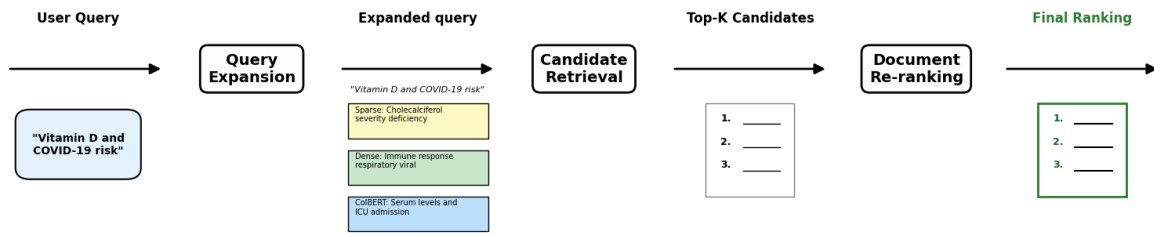


Figure 1: Three-stage pipeline for information retrieval.

Our work is also related to the broader literature on automatic claim verification and, more specifically, to the CheckThat! shared tasks, which have addressed the detection, verification, and analysis of claims in social media across several editions [7, 8, 9]; however, source retrieval for scientific web claims differs from standard verification tasks in an important way: the system is not asked to classify a claim as true or false, but to recover the original scientific paper implicitly referenced in the post and for this reason, the task is closely connected to evidence retrieval, while also requiring robust matching between informal multilingual posts and formal scientific documents.

3. Methodology

Describe the methodology you have adopted, the architecture of your system, your workflow, etc.

We started by implementing a basic pipeline using BM25 as baseline, using Apache Lucene as the underlying framework. We then experimented with different stop lists, filters and weightings to improve the MRR@5 score, used as final score in the CheckThat! Lab. We then proceeded to experiment with multi-stage pipelines based on neural models, which is the final structure of our project.

We implemented a three-stage pipeline (figure 1) for our information retrieval system, consisting of:

- **Query expansion:** We expanded the original query with additional context to bridge the gap between informal and scientific language.
- **Retrieval:** We retrieved relevant documents from the corpus using the expanded query.
- **Reranking:** We reranked the retrieved documents based on their relevance to the original query.

For each of these stages, we experimented with different models, choosing the ones that performed better in relation to the others.

3.1. Query Expansion

The first step in our pipeline is query expansion: given the original query, we generate extra terms and tokens related to it. This serves two purposes: to increase the number of available tokens for a match, and to bridge the gap between the original informal query and the formal academic language used in the document corpus.

Originally, we experimented with expanding the training dataset queries using **TODO: PRIMO MODELLO USATO (bge m3?)** by performing a generative continuation of the query. We then explored different methods for expansion, including Pseudo-Relevance Feedback (PRF) and similarity with static word embeddings using Word2Vec, testing these on both the original and the preliminary expanded queries.

In the final architecture, we settled on using the **TODO: MODELLO ULTIMO USATO** instruction-tuned model to perform generative augmentation. This model was prompted to generate three distinct representations for the given query, tailored to our multi-stage retrieval needs:

- *sparse*: A list of domain-specific keywords and technical synonyms (e.g., scientific nomenclature) designed to maximize lexical overlap and mitigate the vocabulary mismatch in BM25-based retrieval.
- *dense*: A semantically enriched re-phrasing of the claim that provides a broader context, allowing the bi-encoder in the next step of the architecture to produce more robust embeddings in the high-dimensional vector space.
- *ColBERT*: A structured, formal version of the claim (e.g., "Serum levels and ICU admission") that provides coherent token sequences for fine-grained late-interaction re-ranking.

DA CONFERMARE

This multi-view expansion ensures that the downstream hybrid retriever can leverage both high-precision lexical matches and broad semantic overlaps. By generating these representations simultaneously within a single inference step of **TODO: MODELLO ULTIMO USATO**, we maintain a coherent relationship between the different query 'views', effectively providing a unified but multi-faceted context for the retrieval pipeline.

The resulting expanded queries are then fed into the multi-stage retrieval framework described next.

3.2. Retrieval

The retrieval stage is designed as a high-recall candidate generator between query expansion and neural reranking. Its objective is not to make the final relevance decision, but to ensure that the true source paper is included in a sufficiently large candidate set for the subsequent, more expensive cross-encoder stage. For this reason, after experimenting with a Lucene BM25 baseline, BGE-M3 only on dense vectors, the final pipeline uses the hybrid BGE-M3 retriever.

This choice reflects the nature of the task: social media claims often contain informal paraphrases, hashtags, named entities, numbers, and fragments of scientific terminology, so relying only on exact lexical overlap or only on dense semantic similarity would discard useful evidence.

The retriever searches the fixed paper collection provided by the task organizers and named `collection_data.json`.

Each paper is represented at document level by the concatenation of its title and abstract, and the retrieved unit is always the paper pubkey. We did not split papers into smaller chunks because the task requires identifying the source publication rather than retrieving an evidence passage, and using one representation per paper avoids an additional chunk-to-document aggregation step. Although the collection also contains metadata such as venue and authors, the hybrid retriever uses only title and abstract, which are the fields consistently encoded for all documents.

The input queries are the multilingual expanded queries `??????????????????WE NEED TO CONFIRM THIS ??????????????????` produced in the previous stage, stored as a JSON file described in the above section.

The expansion step provides three views of the same information need: an embedding view for semantic matching, a sparse view for lexical and terminology-based evidence, and a `colbert` view for fine-grained token interaction. BGE-M3 is well suited to this setting because it can produce dense vectors, sparse lexical weights, and ColBERT-style multi-vector representations from the same multilingual model [?]. Thus, the retriever can combine broad semantic matching with exact term evidence and token-level matching without switching between unrelated encoders.

Candidate selection is performed in two steps. First, all documents are encoded as dense BGE-M3 vectors and searched with an inner-product FAISS index [1]. This dense stage is used as an efficient broad filter over the whole collection. The default setting retrieves a large pool, `candidate_multiplier × top_k`, with `candidate_multiplier=3` and `top_k=1000`, before the final truncation. This deliberately large candidate set favours recall, which is more important at this stage than precision. Second, the dense candidates are rescored using the sparse and ColBERT representations. The sparse component rewards important overlapping terms, while the ColBERT-style component captures local token correspondences that may be lost in a single dense vector [5].

The final configuration was selected through a sequence of runs recorded during development. We first tuned the lexical baseline by varying the Lucene analyzer, stop lists, stemming strategy, and BM25 searcher variants. These experiments improved the classical baseline but also showed its limitation for this task: even with query expansion and reranking, BM25-based candidate generation remained more dependent on surface overlap than desired. We then tested dense retrieval with BGE models, which improved the semantic coverage of the candidate set, especially for paraphrased or multilingual claims. The best retrieval-only results in the analysis came from the BGE-M3 hybrid variants, confirming the motivation for combining semantic, lexical, and token-interaction evidence in the same first-stage retriever.

Among the hybrid configurations, the run labelled `bge_m3_hybrid_513` was retained because it gave the strongest recall-oriented trade-off among the tested no-rerank retrievers: it used weights 0.5/0.15/0.35 for dense, sparse, and ColBERT scoring, respectively, with multiplier 3 and maximum hybrid length 384. A closely related configuration with a larger ColBERT weight obtained a nearly identical MRR@5 but lower Recall@100, so we preferred the setting that better matched the role of retrieval as a candidate generator. This choice was further confirmed by downstream runs: reranking the candidates from `bge_m3_hybrid_513` consistently outperformed the corresponding BM25-based reranked runs, and increasing the reranked candidate depth from a few hundred documents to 500–1000 documents improved the final recall available to the reranker.

We also experimented with a learned fusion strategy for these hybrid signals. In particular, the experimental test performed includes a LightGBM LambdaMART option that builds features from the dense, sparse, and ColBERT scores, together with normalized score variants and dense-rank features, and then learns how to combine them. However, the retained retrieval configuration uses the simpler linear fusion, which was easier to reproduce and aligned well with the recall-oriented role of this stage. The score used in the final hybrid retriever is therefore:

$$s(q, d) = 0.5 s_{\text{dense}}(q, d) + 0.15 s_{\text{sparse}}(q, d) + 0.35 s_{\text{colbert}}(q, d).$$

The largest weight is assigned to the dense score because it provides the main multilingual semantic signal; the sparse and ColBERT terms add complementary lexical and fine-grained matching evidence. Candidates are sorted by this score in descending order and truncated to the final `top_k`. The retrieval output is a JSON mapping from query id to an ordered list of paper pubkeys. Scores are not exported, because the next stage treats retrieval as a candidate generator and recomputes relevance with a cross-encoder over the returned document identifiers.

3.3. Reranking

3.4. Evaluation

4. Experimental Setup

We conducted about 60 runs on the test split, while a smaller development split was used to inspect intermediate behavior before consolidating the final pipeline; across these runs we varied the query-expansion component, the English analyzer, and the weighting of dense, sparse, and ColBERT signals in the hybrid retriever. We also compared several rerankers, including Nemotron, Qwen3, GTE-ModernBERT, and cascaded reranking strategies, while varying the size of the candidate set passed to the rerankers; this tracking was used to identify representative systems, study the contribution of each component, and support the comparisons discussed in Section 5.

5. Results and Discussion

5.1. Main Results

Table 1 summarizes ten representative runs selected from our experimental campaign; rather than listing the top-scoring runs only, we chose configurations that cover the main families of experiments

Table 1
Representative runs selected from the experimental campaign.

Family	Run label	Key configuration	MRR@5	Recall@100
Analyzer tuning	Analyzer V2	Lexical retrieval with MyEnglishAnalyzer_V2 and BGE-V1 expansion	0.503	0.796
Analyzer tuning	Analyzer V1.12	Lexical retrieval with KStem and custom stoplist variant	0.450	0.783
Retrieval only	Hybrid A	BGE-M3 hybrid retrieval, dense 0.35, sparse 0.15, ColBERT 0.50	0.557	0.837
Retrieval only	Hybrid B	BGE-M3 hybrid retrieval, dense 0.50, sparse 0.15, ColBERT 0.35	0.557	0.839
Single-stage reranking	Nemotron top-400	Hybrid retrieval followed by Nemotron reranking on 400 candidates	0.694	0.893
Single-stage reranking	Nemotron top-1000	Hybrid retrieval followed by Nemotron reranking on 1000 candidates	0.702	0.911
Alternative rerankers	Qwen3-4B	Hybrid retrieval followed by Qwen3 reranking	0.662	0.839
Alternative rerankers	GTE-ModernBERT	Hybrid retrieval followed by GTE-ModernBERT reranking	0.652	0.852
Cascaded reranking	Nemotron + BGE-v2-M3	Nemotron first-stage reranking followed by BGE reranking	0.741	0.893
Cascaded reranking	Nemotron + ModernBERT-FT	Nemotron first-stage reranking followed by fine-tuned ModernBERT reranking	0.766	0.893

explored during the project, namely analyzer tuning, retrieval-only hybrid search, single-stage reranking, alternative rerankers, and cascaded reranking.

The results in Table 1 show a clear progression across the different stages of the project. The best retrieval-only configuration reaches an MRR@5 of 0.557, while the best cascaded system reaches 0.766. The largest individual improvement comes from adding `nvidia/llama-nemotron-rerank-1b-v2` as the first reranking stage, which moves the system from the 0.557 range of the hybrid-only runs to 0.702. A second reranking stage then brings a further gain, with the fine-tuned ModernBERT model producing the strongest final configuration.

5.2. Ablation / Run Analysis

Analyzer tuning: We initially explored stronger English analyzers and several stemming and stop-word variants in order to improve lexical matching between short social-media posts and scientific abstracts. This family peaks at an MRR@5 of 0.503 with `MyEnglishAnalyzer_V2`, while the other analyzer variants remain around the 0.45–0.50 range. These runs were useful in validating the pre-processing choices, but they also showed that text normalization alone is not enough to bridge the semantic gap between informal claims and academic documents. Analyzer tuning yields measurable improvements, but its impact is limited when compared with semantic retrieval and neural reranking.

Hybrid retrieval without reranking: A shift from analyzer-focused experiments to hybrid retrieval leads to a substantially larger improvement; the two best retrieval-only systems, reported in Table 1 as Hybrid A and Hybrid B, both reach an MRR@5 of about 0.557, with recall@100 between 0.837 and 0.839. The difference between them lies in the balance between the dense and ColBERT components, while the sparse contribution is kept fixed. This indicates that combining dense, sparse, and ColBERT-style signals already produces a strong candidate set before any reranking is applied; at the same time the very small difference between the two configurations suggests that, once the hybrid formulation is in place, the exact internal weighting matters less than the transition away from purely lexical retrieval. Hybrid retrieval is the first major step change in effectiveness and provides the strongest retrieval-only foundation for the full pipeline.

Single-stage reranking with Nemotron: Adding `nvidia/llama-nemotron-rerank-1b-v2` on top of the hybrid candidates produces the largest isolated gain in MRR@5 ; the representative Nemotron runs increase from the 0.557 range of the retrieval-only systems to 0.694 and 0.702, while recall@100 also rises from about 0.839 to as high as 0.911. We also observe a limited but consistent benefit from increasing the number of candidates exposed to the reranker: moving from the top-400 to the top-1000 setting slightly improves both MRR@5 and recall@100 ; this suggests that the reranker can exploit a broader candidate pool without being significantly affected by additional noise. The first reranking stage is the most impactful component of the pipeline, and a larger candidate pool brings further, although diminishing, gains.

Alternative rerankers: To verify that the gains were not tied to a single model, we also evaluated other rerankers, including `Qwen3-reranker-4B` and `Alibaba-NLP/gte-reranker-modernbert-base`; both models clearly outperform the retrieval-only baseline in terms of MRR@5 , reaching values of 0.662 and 0.652, respectively. However, they remain below the Nemotron runs, especially on MRR@5 , which suggests weaker ranking quality in the first positions of the final list; even so these experiments confirm that the hybrid retriever produces candidate sets that can be improved by different neural rerankers. Alternative rerankers validate the overall pipeline design, but Nemotron is the strongest first-stage reranker among the models we tested.

Cascaded reranking: Our best results are obtained by applying a second reranking stage to the candidates already filtered by Nemotron; the cascade based on `BAAI/bge-reranker-v2-m3` reaches an MRR@5 of 0.741, while the best overall system, based on a fine-tuned version of `answerdotai/ModernBERT-large`, reaches 0.766. Both cascaded runs keep recall@100 at 0.893, which is slightly lower than the Nemotron top-1000 configuration but still substantially higher than the retrieval-only systems; this indicates that the second reranker mainly improves the ordering of already strong candidates rather than increasing the breadth of the candidate pool. Cascaded reranking is the best strategy to maximize early precision, and the fine-tuned ModernBERT reranker provides the strongest final ranking.

6. Conclusions and Future Work

Our experiments show that the strongest improvement comes from neural reranking, especially when a hybrid retriever is followed by `nvidia/llama-nemotron-rerank-1b-v2` and then refined by a second reranking stage based on fine-tuned ModernBERT. In contrast, analyzer tuning and other lexical adjustments produced only limited gains when compared with the impact of hybrid retrieval and multi-stage reranking. Future work will focus on improving the multilingual side of the pipeline and on further refining the second-stage reranker.

7. Group Members Contribution

Fabio Baldan has contributed to this project by:

1. Helping designing and implementing the initial BM25-basic pipeline;
2. Writing and revising some chapters of the documentation based on the actual code and on the concepts learned during the project development;
3. Tracking and reviewing the annotation-part for the runs with various systems;
4. Trying various models for the re-rank part to enhance the MRR@5 ;
5. Implementing a custom evaluator to focus on metrics that matter the most;
6. Reviewing and improving the query-expansion;
7. Fine-tuning the re-reranker to improve the MRR@5 and assure an advantage if compared with the default one.

Davide Donati has contributed to this project by:

1. Helping designing and implementing the initial BM25 basic pipeline;
2. Implemented a Boolean Query system for the searcher, improving the performance of the retrieval;
3. Writing part of the docs and helping other members by reviewing their writes;
4. Keeping track of the results of the systems that he tried;
5. Improving the multi-lingual query-expansion.

Alvise Garberino has contributed to this project by:

1. Helping designing and implementing the initial BM25-basic pipeline;
2. Helping and searching the best models and techniques to implement a re-ranker model;
3. Researching and implementing functionalities for the re-ranker such as data augmentation and data-distillation;
4. Writing some chapters of the documentation providing comprehensive citations and references to the academic material used in the project;
5. Keeping track of the results of the systems that he tried.

Giancarlo Padoan has contributed to this project by:

1. Designing, implementing and fixing the initial basic pipeline;
2. Writing and correcting some chapters of the documentation based on the actual code and on the notions learned while building the project;
3. Designing and implementing an hybrid-retrieval system to enhance recall, which is the core of the final architecture;
4. Trying various models for the embedding-representation part to enhance the MRR@5, ;
5. Implementing the first version of query-expansion functionality, and improving it by testing different techniques;
6. Tested different scoring function for combining the hybrid-retrieval scores.
7. Testing various cross-encoder models for the re-ranking stage of the pipeline.
8. Keeping track of the results of the systems that he tried.

Marco Tessari has contributed to this project by:

1. Helping designing and implementing the initial BM25-basic pipeline;
2. Improved the performance of the English Analyzer (text-processing, stemmer, filters and stopwords);
3. Writing part of the docs and helping other members by reviewing their writes;
4. Keeping track of the results of the systems that he tried;
5. Helping and searching the best models and techniques to implement an embedding model;
6. Experimenting with word2vec static embeddings for query expansion
7. Helping to find the best weights to enhance recall in the hybrid-retrieval system.

8. Misc [TO BE REMOVED]

INTRODUCTION.TEX E RELATED.TEX SONO DA FINIRE QUANDO AVREMO I RISULTATI COMPLETI (VEROSIMILMENTE QUALCHE GIORNO PRIMA DELLA CONSEGNA) NON CANCELLARE QUESTA FRASE E NON MODIFICARE QUEI FILE (SE NON PER AGGIUNGERE COSE UTILI) SENNÒ MI DIMENTICO DI FARLO E PRENDIAMO 5/20 GRAZIE :) <3

8.1. Tex Files

Put your $\text{\LaTeX}$ files into the `section` folder as shown in the examples above.



Figure 2: 1907 Franklin Model D roadster. Photograph by Harris & Ewing, Inc. [Public domain], via Wikimedia Commons. (<https://goo.gl/VLCRBB>).

Table 2

Frequency of Special Characters

Non-English or Math	Frequency	Comments
Ø	1 in 1,000	For Swedish names
π	1 in 5	Common in math
\$	4 in 5	Used in business
Ψ_1^2	1 in 40,000	Unexplained usage

8.2. Figures

Put your figures into the `figure` folder and put the caption under the figure. Example of reference to Figure 2.

8.3. Tables

Put the caption above the table. Example of reference to Table 2.

See the `booktab` packaged documentation for further options.

8.4. Bibliography

Example of citations:

- name: Salton [10]
- parenthesis: [10]

An initial list of references is provided in the files `bibliography.bib` and `proceedings.bib` that you can expand.

See the `natbib` packaged documentation for further options.

8.5. Acronyms

Use the

```
\ac{acronym}
```

command to insert acronyms, eg. *Average Precision (AP)*. The command will expand the acronym the first time it is used.

An initial list of acronyms is provided in the file `acronyms.tex` that you can expand.

See the acronym packaged documentation for further options.

References

- [1] Faiss, Faiss documentation, <https://faiss.ai/>, ????. Accessed: 01-05-2026 (DD-MM-YYYY, italian date format).
- [2] S. E. Robertson, U. Zaragoza, The Probabilistic Relevance Framework: BM25 and Beyond, *Foundations and Trends in Information Retrieval (FnTIR)* 3 (2009) 333–389.
- [3] S. Marchesin, A. Purpura, G. Silvello, Focal elements of neural information retrieval models. An outlook through a reproducibility study, *Information Processing & Management* 57 (2020) 102109:1–102109:29.
- [4] V. Karpukhin, B. Oguz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, W.-t. Yih, Dense Passage Retrieval for Open-Domain Question Answering, in: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Association for Computational Linguistics, Online, 2020, pp. 6769–6781. URL: <https://aclanthology.org/2020.emnlp-main.550/>. doi:10.18653/v1/2020.emnlp-main.550.
- [5] O. Khattab, M. Zaharia, ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT, in: *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2020)*, ACM Press, New York, USA, 2020, pp. 39–48. URL: <https://doi.org/10.1145/3397271.3401075>. doi:10.1145/3397271.3401075.
- [6] R. Nogueira, Z. Jiang, R. Pradeep, J. Lin, Document Ranking with a Pretrained Sequence-to-Sequence Model, in: *Findings of the Association for Computational Linguistics: EMNLP 2020*, Association for Computational Linguistics, Online, 2020, pp. 708–718. URL: <https://aclanthology.org/2020.findings-emnlp.63/>. doi:10.18653/v1/2020.findings-emnlp.63.
- [7] T. Elsayed, P. Nakov, A. Barrón-Cedeño, M. Hasanain, R. Suwaileh, G. Da San Martino, P. Atanasova, Overview of the CLEF-2019 CheckThat!: Automatic Identification and Verification of Claims, in: F. Crestani, M. Braschler, J. Savoy, A. Rauber, H. Müller, D. E. Losada, G. Heinatz Bürki, L. Cappellato, N. Ferro (Eds.), *Experimental IR Meets Multilinguality, Multimodality, and Interaction. Proceedings of the Tenth International Conference of the CLEF Association (CLEF 2019)*, Lecture Notes in Computer Science (LNCS) 11696, Springer, Heidelberg, Germany, 2019, pp. 301–321.
- [8] A. Barrón-Cedeño, T. Elsayed, P. Nakov, G. Da San Martino, M. Hasanain, R. Suwaileh, F. Haouari, N. Babulkov, B. Hamdan, A. Nikolov, S. Shaar, Z. Sheikh Ali, Overview of CheckThat! 2020: Automatic Identification and Verification of Claims in Social Media, in: A. Arampatzis, E. Kanoulas, T. Tsikrika, S. Vrochidis, H. Joho, C. Lioma, K. Eickhoff, A. Névél, L. Cappellato, N. Ferro (Eds.), *Experimental IR Meets Multilinguality, Multimodality, and Interaction. Proceedings of the Eleventh International Conference of the CLEF Association (CLEF 2020)*, Lecture Notes in Computer Science (LNCS) 12260, Springer, Heidelberg, Germany, 2020, pp. 215–236.
- [9] P. Nakov, G. Da San Martino, T. Elsayed, A. Barrón-Cedeño, R. Míguez, S. Shaar, F. Alam, F. Haouari, M. Hasanain, W. Mansour, B. Hamdan, Z. S. Ali, N. Babulkov, A. Nikolov, G. K. Shahi, J. M. Struß, T. Mandl, M. Kutlu, Y. S. Kartal, Overview of the CLEF-2021 CheckThat! Lab on Detecting Check-Worthy Claims, Previously Fact-Checked Claims, and Fake News, in: K. S. Candan, B. Ionescu, L. Goeuriot, B. Larsen, H. Müller, A. Joly, M. Maistro, F. Piroi, G. Faggioli, N. Ferro (Eds.), *Experimental IR Meets Multilinguality, Multimodality, and Interaction. Proceedings of the Twelfth*

International Conference of the CLEF Association (CLEF 2021), Lecture Notes in Computer Science (LNCS) 12880, Springer, Heidelberg, Germany, 2021, pp. 264–291.

[10] G. Salton, Automatic Information Organization and Retrieval., McGraw-Hill, New York, USA, 1968.